Real portraits

Source	Scale / notes	License / access	Role
FFHQ ✔	70k PNG, 1024^2, Flickr	Individual images carry CC BY 2.0, CC BY-NC 2.0, PDM 1.0, CC0 or US Gov Works licenses; the dataset (metadata, script, docs) is CC BY-NC-SA 4.0 from NVIDIA. Flickr owners can opt out of the dataset.	Real train. It is the training set for StyleGAN, so treat it as a contamination risk (see §5).
CelebA / CelebA-HQ / CelebAMask-HQ	200k / 30k; celebrity identities	Non-commercial research only; request form	Real, identity-labelled. Celebrity-heavy, so a strong domain bias.
FairFace	~100k, balanced race/age/gender	CC BY 4.0 (verify)	Real, demographic eval slices
Casual Conversations (Meta v1/v2)	Consented, self-reported attributes	Request-based, restricted use	Real eval-only, fairness slices
Open Images	Large, in-the-	CC BY 2 ages, CC BY 4.0	Real, varied

(Meta v1/v2)	attributes		
Open Images (face crops)	Large, in-the-wild	CC BY 2.0 images, CC BY 4.0 annotations (verify per image)	Real, varied capture conditions
LFW, UTKFace, WIDER FACE	Small to medium	Research-only or non-commercial, unclear provenance	Eval only, if at all
Wikimedia Commons, Unsplash, Pexels	Fresh, out-of-distribution photos	Per-image licenses, and personality-rights issues for identifiable people	Held-out real test set, where you control curation
Own capture with consent	Phone and DSLR variety	You define the license	Highest-value real test set, since no generator has seen it juheapi

Avoid anything withdrawn or ethically contested (MS-Celeb-1M, VGGFace2, and similar). Many are pulled or access-restricted, and BIPA, ↓ R Art. 9 and CCPA make biometric data

		CelebAMask-HQ terms. Verify release status.
DFFD (MSU), 100K-Faces, iFakeFaceDB, ForenSynths, GenImage, Synthbuster, WildFake	GAN-era and early diffusion coverage	Mixed academic licenses; re-verify. Most are not face-specific.
ThisPersonDoesNotExist-style dumps	StyleGAN2 only	Provenance weak. Prefer generating your own.
Self-generated	Controlled, fully labelled	See below huggingface.co +4

Tier	Examples	Can you train a commercial detector on it?	Redistribute your derived dataset?
A: open	CC0/PDM subsets of FFHQ, own capture	Yes	Yes
B: attribution or share-alike	CC BY images, FFHQ CC BY 2.0 subset	Yes, with attribution manifest	Yes, with per-image attribution
C: non-commercial	FFHQ metadata (BY-NC-SA), CelebA, CC BY-NC images	Research only	Share-alike or no
D: request-gated / eval-only	Casual Conversations, DF40	Per agreement	No
E: generator-terms-dependent	Midjourney, DALL·E, FLUX dev, OpenRAIL-M models	Read the terms. Many bar using outputs to train competing models, and OpenRAIL has use restrictions.	Often no

Split	Synthetic content	Real content	Purpose
train	~60–70% of families (e.g. ProGAN, StyleGAN2, SD1.5, SDXL, selected swaps)	FFHQ (BY/PDM/CC0), Open Images, other A/B sources	Fit
val	Unseen checkpoints from *train* families, plus group-disjoint images	Group-disjoint, same sources	Tuning and early stopping. No test families here.
test_in_family	Held-out checkpoints, seeds and prompts of train families	Group-disjoint	Sanity ceiling
test_heldout_gen	Whole generators unseen but in a seen family (e.g. SD 2.1, PixArt if DiT is in train)	Disjoint sources	Moderate generalization
test_heldout_family	A whole family excluded from train (e.g. all DiT/flow, or all closed API)	Disjoint sources	Hard generalization
test_temporal	Generators released after the training cutoff date	Fresh captures	Realistic deployment

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test_temporal	Generators released after the training cutoff date, refreshed on a schedule	Fresh captures	Realistic deployment drift
test_real_ood	Mixed	Own-capture and Wikimedia/Unsplash reals, never used in any training	Guards against "FFHQ-looking = real" shortcuts

Protocol and reporting

- **Leave-one-family-out:** rotate each family as the held-out test, and report a matrix, not a single score.
- **Fixed operating point:** pick the threshold on val, then report TPR at 1% and 0.1% FPR on every test split, with AUC alongside. Never retune thresholds on test splits.
- **Per-generator breakdown:** AUC and detection rate per `generator_id`, plus worst-